as we mentioned in the definition of principle solution. If we take the principal solution in the interval $[0, 2\pi]$, then the principal solution is $\theta = \frac{4\pi}{3}$ and the general solution is

$$\theta = n\pi + (-1)^n \left(\frac{4\pi}{3}\right), n \in \mathbb{Z}, \frac{4\pi}{3} \in [0, 2\pi] \quad \text{... (ii)}$$

From (ii), for $n = 0, -1, 1, -2, 2, \ldots$
the corresponding solutions are $\frac{4\pi}{3}, \frac{-7\pi}{3}, \frac{-\pi}{3}, \frac{-2\pi}{3}, \frac{10\pi}{3}, \ldots$
From (i), for $n = 0, -1, 1, -2, 2, \ldots$
the corresponding solutions are $\frac{-\pi}{3}, \frac{-2\pi}{3}, \frac{4\pi}{3}, \frac{-7\pi}{3}, \frac{5\pi}{3}, \ldots$

In both the cases, we get the same set of solutions, but in different order. Thus, we have justified that the principal solution can be taken either in $[0, 2\pi]$ or in $[-\pi, \pi]$.

Example 3.44 Find the general solution of
(i) $\sec\theta = -2$ (ii) $\tan\theta = \sqrt{3}$

Solution:

(i) $\sec\theta = -2$

$\sec\theta = -2 \Rightarrow \cos\theta = -\frac{1}{2}$

We know that the general solution of $\cos\theta = \cos\alpha, \alpha \in [0, \pi]$, is $\theta = 2n\pi \pm \alpha, n \in \mathbb{Z}$

Let us find $\alpha \in [0, \pi]$ such that $\cos\alpha = -\frac{1}{2} = \cos\left(\pi - \frac{\pi}{3}\right) = \cos\frac{2\pi}{3}$.

so, $\alpha = \frac{2\pi}{3}$.

Thus, the general solution is $\theta = 2n\pi \pm \frac{2\pi}{3}, n \in \mathbb{Z}$.

(ii) $\tan\theta = \sqrt{3}$

$\tan\theta = \sqrt{3} = \tan\frac{\pi}{3}$

We know that the general solution of $\tan\theta = \tan\alpha, \alpha \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ is $\theta = n\pi + \alpha, n \in \mathbb{Z}$

Thus, $\theta = n\pi + \frac{\pi}{3}, n \in \mathbb{Z}$, is the general solution.

Example 3.45 Solve $3\cos^2\theta = \sin^2\theta$

Solution:

$3\cos^2\theta = 1 - \cos^2\theta \Rightarrow \cos^2\theta = \frac{1}{4}$

$\frac{\cos 2\theta + 1}{2} = \frac{1}{4} \Rightarrow \cos 2\theta = -\frac{1}{2} = \cos\left(\pi - \frac{\pi}{3}\right) = \cos\left(\frac{2\pi}{3}\right)$

$2\theta = 2n\pi \pm \frac{2\pi}{3}, n \in \mathbb{Z} \Rightarrow \theta = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$.

Try to solve by writing $\tan^2\theta = 3$.